

Telco Customer Churn Prediction

ICT6001C — AI Programme | Week 1 Capstone

Group 3

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Kaggle Dataset : [Predict Customer Churn | Kaggle](#)

Repo : <https://github.com/kinwahlai/ICT6001C>

Video: <https://youtu.be/s6XiefyZ9nQ>

Executive Summary

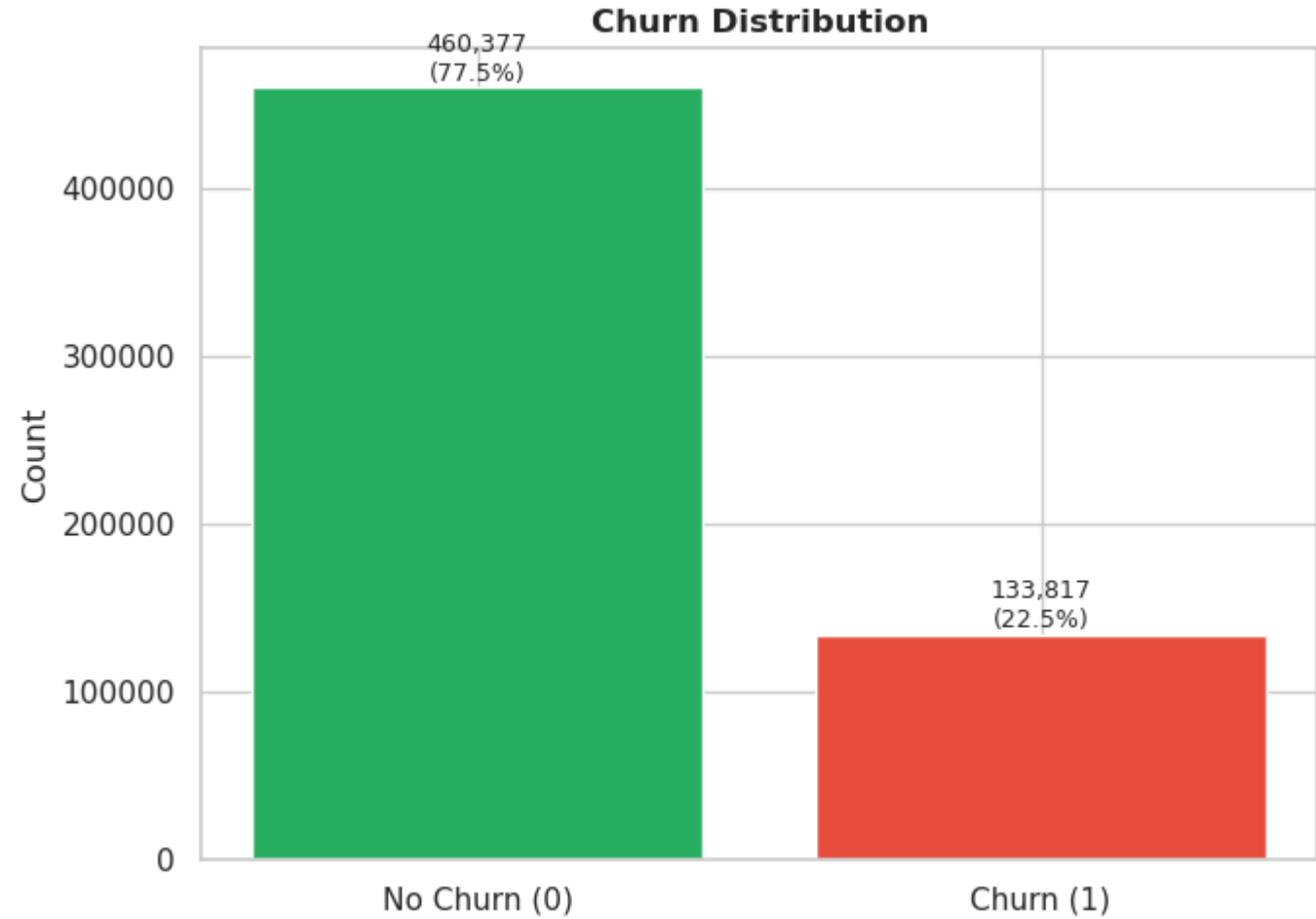
- ◆ Dataset: 594,194 Telco customers | Churn rate ~22.5% | Class imbalance handled with SMOTE
- ◆ 3 model families compared via 5-fold stratified CV (ROC-AUC primary, Recall secondary)
- ◆ Champion: XGBoost (Boosting) — highest CV ROC-AUC
- ◆ Champion tuned via GridSearchCV (champion-only, ≤50-config budget, 100,000 subsampling); CV **ROC-AUC** 0.9109 ± 0.0011
- ◆ Hold-out (evaluated once): ROC-AUC 0.9143 · Recall 0.8083 at threshold 0.61
- ◆ Threshold 0.61 = highest cut still holding ≥80% recall; FN far costlier than FP
- ◆ Tier retention offers by predicted churn probability — concentrate premium spend on highest-risk customers.

Data Engineering & EDA [Part A (§1)]

CRISP-DM: Data Preparation | 80/20 stratified split | X_{test} locked until §4

Data Preparation

- ◆ No missing values; TotalCharges coerced from blank strings.
- ◆ Churn rate ~22.5% — moderate imbalance addressed by SMOTE inside pipeline.



Data Engineering & EDA [Part A (§1)]

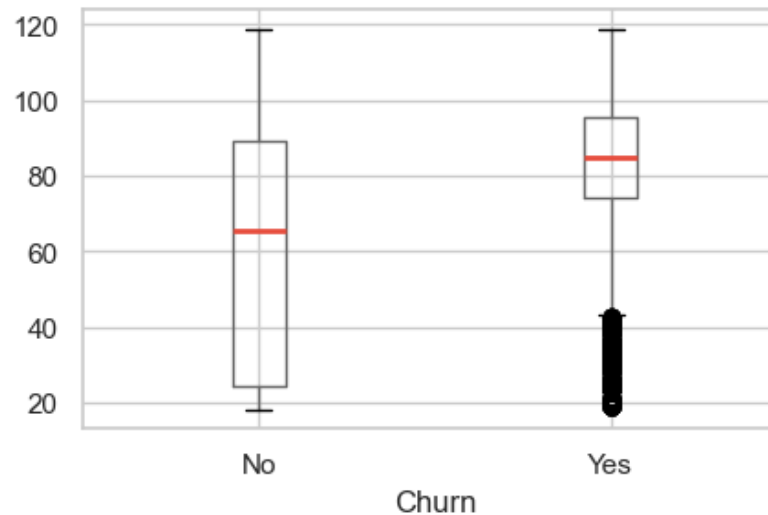
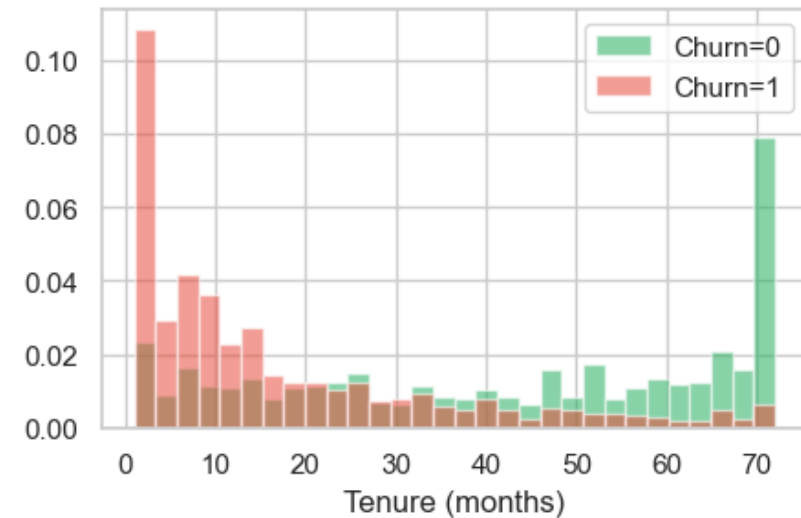
CRISP-DM: Data Preparation | 80/20 stratified split | X_{test} locked until §4

EDA Insights

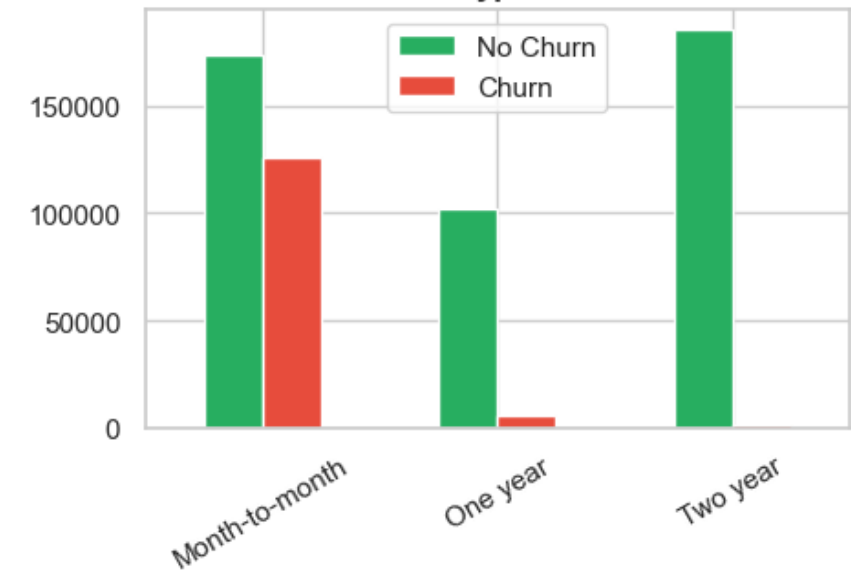
- ◆ Tenure: New customers (<12 months) highest churn risk — focus retention on year 1.
- ◆ MonthlyCharges: Churners pay ~\$15–20/month more (pricing sensitivity).
- ◆ Contract type: Month-to-month churn ~42%; two-year contracts <3%.

Key Drivers of Churn (EDA)

Tenure vs Churn

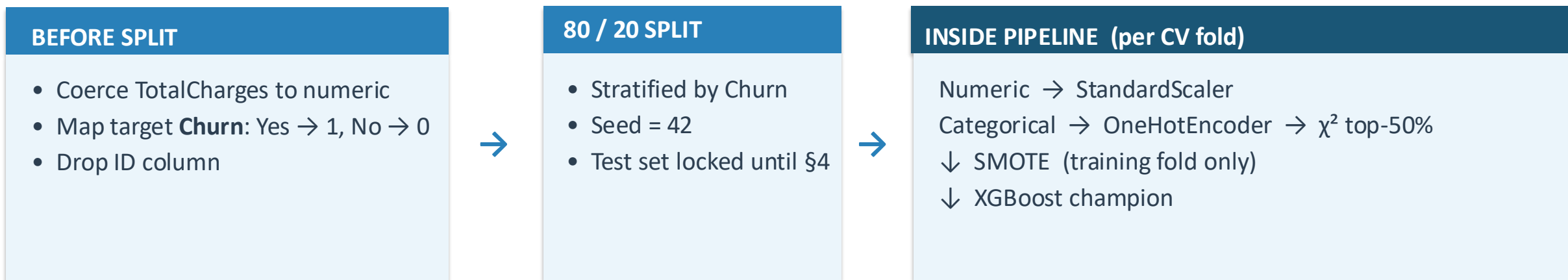


Contract Type vs Churn



Leakage-Safe Pipeline Architecture [Part A (§1)]

CRISP-DM: Data Preparation | All transforms encapsulated inside sklearn Pipeline



Design Decisions

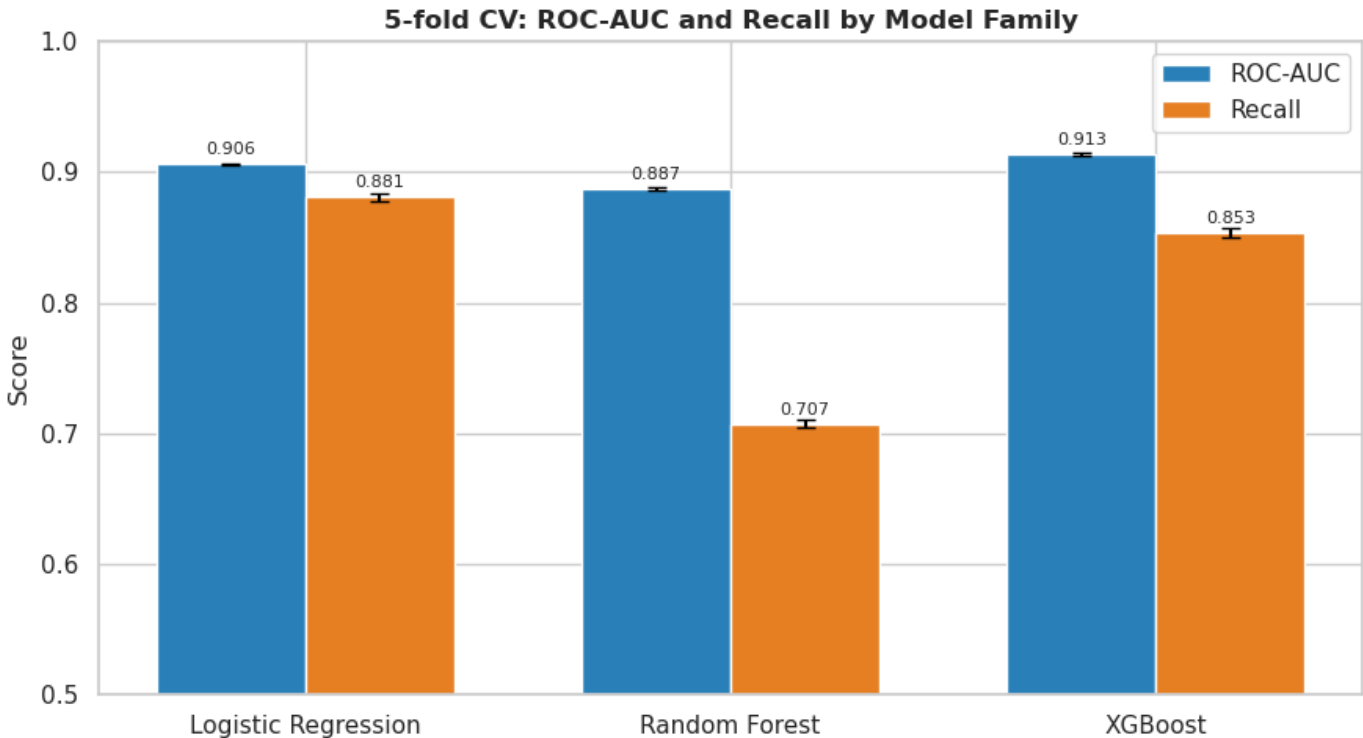
- ◆ Explicitly converting string labels for target **Churn** to define the positive and negative classes.
- ◆ Feature Scaling for Numeric features using StandardScaler to ensure larger ranges (e.g. TotalCharges) do not dominate smaller ones (e.g. Tenure)
- ◆ Dimensionality reduction using chi-squared (χ^2) feature selection to keep the top-50% of encoded categories most associated with churn.
- ◆ Handle imbalanced classes using SMOTE which runs inside the pipeline — oversampling touches training folds only, never validation/test.
- ◆ Using Pipeline ensures that every transform is fit on the training fold only — the test set is untouched until §4.

Champion Model Selection [Part B (§2)]

5-fold stratified CV at representative defaults | ROC-AUC primary, Recall secondary

Family	Model	Rationale	ROC-AUC (mean±std)	Recall (mean±std)
Linear	Logistic Regression	Interpretable baseline; linear decision boundary	0.9059±0.0011	0.8808±0.0027
Bagging	Random Forest	Variance reduction; built-in feature importance	0.8872±0.0013	0.7072±0.0024
Boosting	XGBoost	Sequentially corrects errors; strong on tabular data	0.9134±0.0012	0.8534±0.0038

- ◆ **Primary Metric [ROC-AUC]:** threshold-independent ranking quality — fair comparison across all families.
- ◆ **Secondary Metric [Recall]:** tracks business cost of missed churners (False Negative). threshold is set in §4.



Champion Justification: XGBoost [Part B (§2)]

Selected by highest 5-fold CV ROC-AUC at default params

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Champion Model: XGBoost (Boosting family)

CV ROC-AUC: 0.9134±0.0012 CV Recall: 0.8534±0.0038

Why this champion?

- ◆ XGBoost achieved the highest mean CV ROC-AUC across all 3 families.
- ◆ Low standard deviation across folds → consistent performance, low overfitting risk.
- ◆ ROC-AUC is threshold-independent — a fair selection criterion before §4 threshold shift.
- ◆ Boosting family: sequentially corrects errors — strongest on tabular, imbalanced data.
- ◆ Provides strong baseline for GridSearchCV tuning in §3 (champion-only grid search).

Hyperparameter Tuning for Champion Model XGBoost [Part C (§3)]

GridSearchCV on champion only | ≤ 50 candidate-config cap

Decision	Choice	Rationale
Search method	GridSearchCV	Exhaustive over small grids; reproducible
Scope	Champion only	One champion declared in §2; tune alone
Iterations cap	≤ 50 candidate configs	Champion grid with 48 configs (well within cap)
CV folds	5-fold stratified	Validation repeats, not search iterations
Tune dataset	100,000 subsample	4–5× faster; refit on full 475,000 rows
refit metric	roc_auc	Threshold-independent; recall tuned in §4

Tuning Parameters

```
"XGBoost": {  
  "classifier__max_depth": [3, 4, 6, 8],  
  "classifier__learning_rate": [0.1, 0.3, 0.5, 1],  
  "classifier__n_estimators": [100, 200, 300],  
},
```

Tuning Outcome

Champion: XGBoost

Best params: learning_rate=0.1, max_depth=3, n_estimators=300

CV ROC-AUC: 0.9109±0.0011

CV Recall: 0.8535±0.0019

Refit best config on full 475,000 rows train set

Final Model Stability [Part B (§2)] → [Part D (§4)]

CV cross-validated scores (train) + Hold-out evaluation (test, accessed once)

5-fold CV Scores for XGBoost

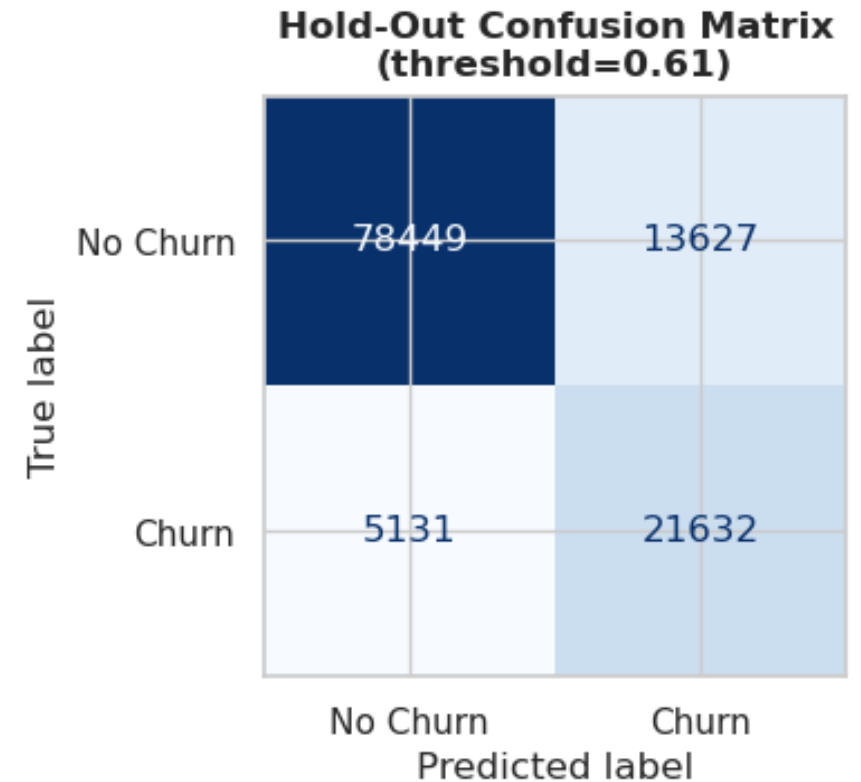
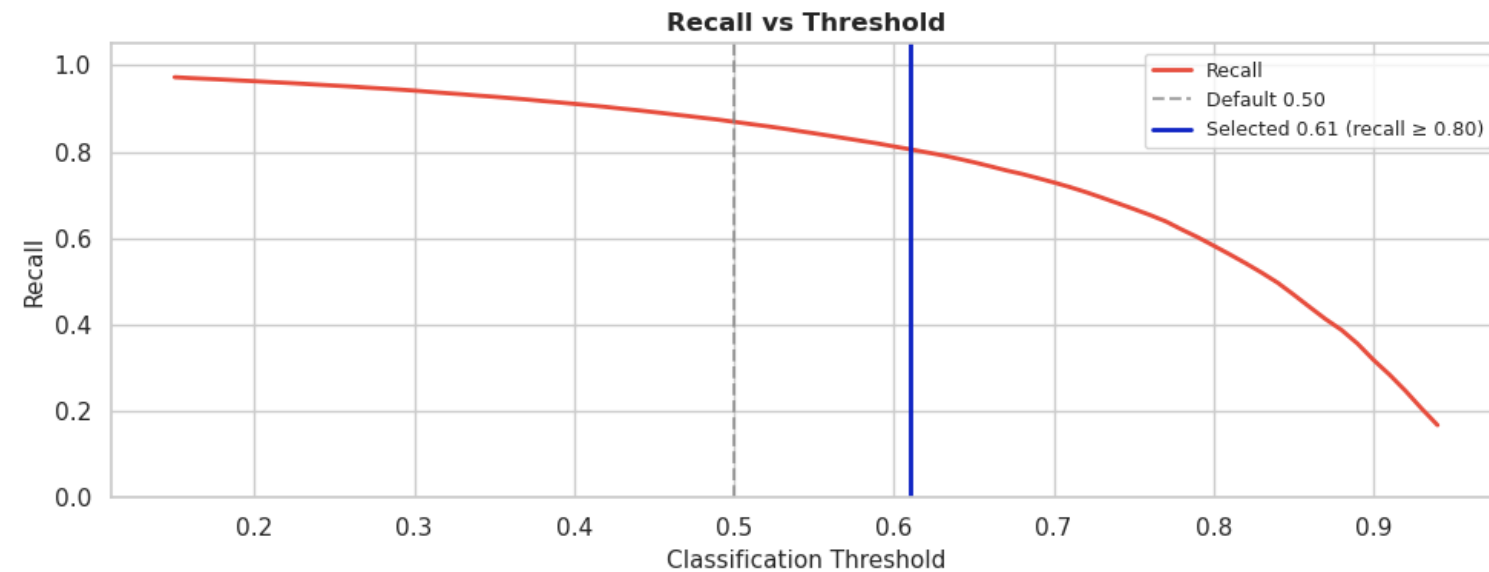
Model Test	ROC-AUC	Recall	Dataset Size
Before Tuning	0.9134±0.0012	0.8534±0.0038	475,000
Before Tuning	0.9078±0.0008	0.8236±0.0053	100,000
After Tuning	Unable to converge due to huge dataset (475,000 rows) with GridSearchCV (48 candidate configs) 48 candidate configs X 5-fold CV = 240 fits		475,000
After Tuning	0.9109±0.0011	0.8535±0.0019	100,000

Final Model Stability [Part B (\$2)] → [Part D (\$4)]

CV cross-validated scores (train) + Hold-out evaluation (test, accessed once)

Hold-Out Test Metrics (threshold = 0.61)

Metric	Value
ROC-AUC	0.9143
Recall	0.8083



Business Decision Reasoning [Part D (\$4)]

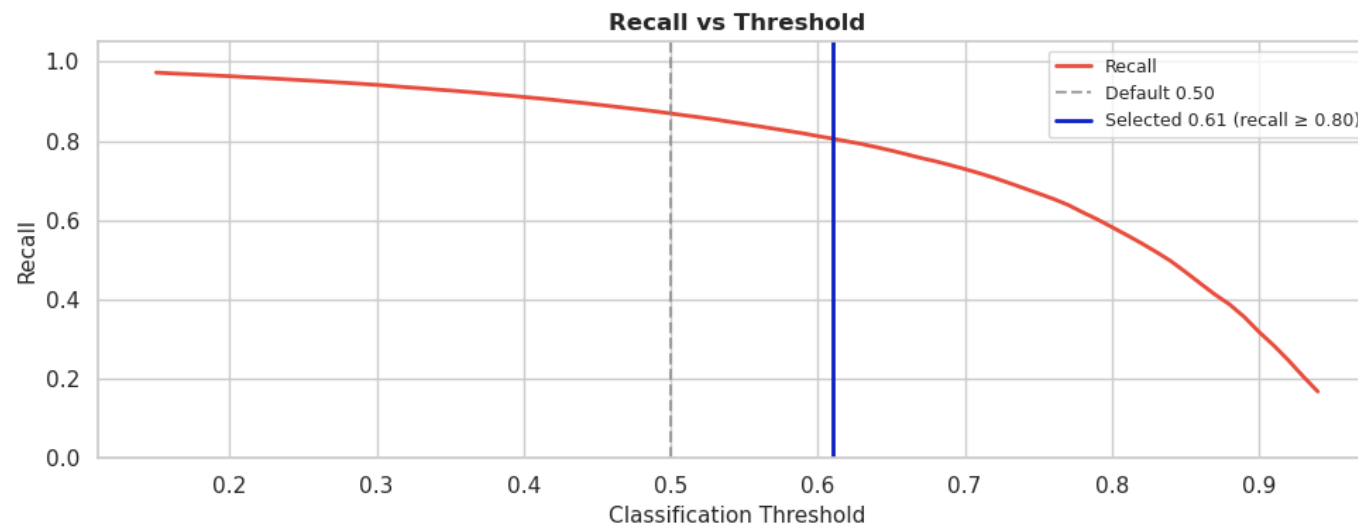
CRISP-DM: Deployment | Asymmetric error costs | Threshold shift

Asymmetric Error Costs

Error Type	Consequence	Severity
False Negative — miss a churner	Customer leaves uncontacted → lost lifetime value	CRITICAL
False Positive — flag loyal customer	Wasted retention offer — comparatively cheap, recoverable	Moderate

Threshold Decision

- ◆ FN far costlier than FP → prioritise recall; require recall $\geq 80\%$.
- ◆ Among thresholds meeting the floor, 0.61 is the highest — limits false positives.
- ◆ At threshold 0.61: Recall 0.8083 on hold-out — 81% of real churners flagged for intervention.
- ◆ Tier retention offers by predicted churn probability — concentrate premium spend on highest-risk customers.



Individual Contributions

Name	Student ID
YEO XIU JUAN	2570472
LIM KHOON SENG	2570459
LAI KIN WAH	2570461

Work Area	Primary Owner	Secondary Support
Part A: EDA & Data Quality	LAI KIN WAH	YEO XIU JUAN
Part A: Pipeline Engineering	YEO XIU JUAN	LIM KHOON SENG
Part B: Model Selection	YEO XIU JUAN	LIM KHOON SENG
Part C: Hyperparameter Tuning	LIM KHOON SENG	LAI KIN WAH
Part D: Business Decision	LIM KHOON SENG	LAI KIN WAH
Report & Slides	LAI KIN WAH	LIM KHOON SENG

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